

Motor driver

DO

OUTPUT

MOVEMENT

What does it do?

A motor driver lets a low-power GPIO control a motor that needs more current.

The L293D contains H-bridges that can send current through a motor in either direction.

Two input pins choose forwards, backwards or stop, while the chip handles the motor current.

Uses

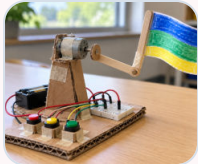


Electric car window

DETECT
Read window
switch

DECIDE
Can it move
safely?

DO
Drive
requested
direction

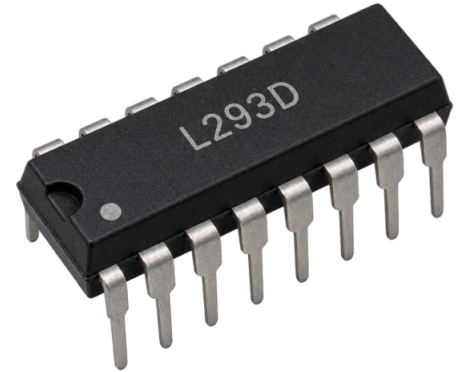


Tiny flag waver

DETECT
Count score-
button presses

DECIDE
Has the
target been
reached?

DO
Wave the flag



Notch
Identifies
pin 1

**IN1 and
IN2**
Direction
controls

**OUT1
and
OUT2**
Connect
to the
motor

Enable
Held high
in this
circuit

Code

● Raspberry Pi

```
from gpiozero import Motor
motor = Motor(forward=17, backward=18)
motor.forward()
```

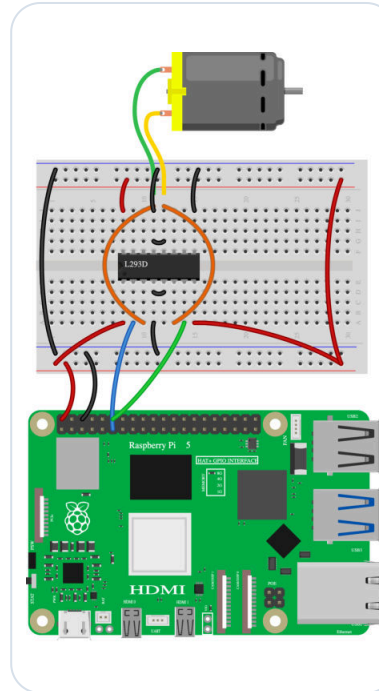
● Raspberry Pi Pico

```
from picozero import Motor
motor = Motor(14, 15)
motor.forward()
```

This card powers one small 5 V motor from the board's 5 V rail. Use a separate motor supply for larger motors, with the grounds joined.

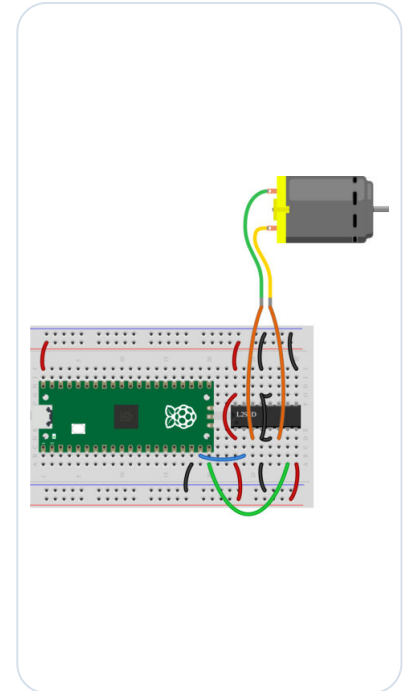
Wiring

RASPBERRY PI 5



GPIO17 → IN1, **GPIO18** → IN2; OUT1 and OUT2 → motor

PICO



GP14 → IN1, **GP15** → IN2; OUT1 and OUT2 → motor